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Development of High-Performance Completion and Stimulation Techniques for 200°C Extreme High-Temperature Reservoirs

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Abstract

The Kunteyi block is one kind of an ultra-deep, ultra-high pressure, and ultra-high temperature bedrock gas reservoir in the Qaidam Basin. The extreme reservoir characteristics of 7000m+ and 200°C pose significant challenges to well completion and stimulation. Three major challenges are: 1) lack of experience in well completion operations for ultra-deep reservoirs at 200°C; 2) what kind of well completion and Stimulation technology can meet the requirements for maximizing well productivity; 3) whether the performance of tools and fluids at 200°C meet the process requirements.

This paper introduces four well completion and stimulation technologies for 200°C deep wells, and provides field application scenarios and experiences of different types of well completion and stimulation. The well completion can be divided into four stages: First Stage: For the first well in this block, the THT completion packer was used after combined logging and perforation operations. Second Stage: The integrated acidizing completion string was adopted, which comprises single/multi-layer acidizing strings combined with THT permanent packers. Third Stage: After tubing perforation, an acidizing string with Y531 packers was used for well completion, followed by pilot production testing. Fourth Stage: For the ultra-deep, high-angle wells in this area, a combined plugging and perforation operation with casing fracturing stimulation was employed for well completion. Furthermore, the corresponding perforation technologies, types of packers, staged plug types, pressure-bearing calculations for hangers, and the performance of corresponding well completion tools for each of these four processes are introduced in detail.

The reverse design and forward implementation method significantly reduce downhole accidents such as wellbore blockage and pipe sand burial. For a 7-inch + 5-inch wellbore, the selection of a THT packer

corresponding to the 7-inch casing is optimized, with performance requirements meeting 103.4MPa pressure resistance and temperature resistance of 232°C. Field tests have shown that a downhole hanger with a pressure-bearing capacity of 105MPa can meet the stimulation treatment requirements, but there are certain risks involved. For subsequent wells, theoretical calculations have optimized the selection of a hanger with a pressure-bearing capacity of 140MPa. The perforating gun is preferably an ultra-high temperature 89-gun, using a second-generation-plus bullet type, which has a temperature resistance of 210°C/48h and a pressure resistance of 175MPa. Moreover, the ultra-high temperature fracturing fluid has achieved a 74% drag reduction, and the ultra-high temperature acid fluid has reduced the construction pressure by up to 15.4 MPa.

This paper presents a novel integrated design approach for completion and stimulation for ultra-deep and ultra-high temperature wells. Four different completion techniques and treatment experiences are shared, providing referenceable insights and practical experience for optimizing well completion and reservoir stimulation in ultra-deep reservoirs at 200°C.

Keywords: Ultra-Deep and High-Temperature Reservoir, Well completion, Stimulation, challenge, Experience sharing

Introduction

With global oil and gas exploration and development continuously advancing into deep and ultra-deep formations (temperatures ≥150°C), ultra-high temperature (UHT) reservoirs have become a core focus for increasing reserves and production both domestically and internationally, as well as for geothermal development. These reservoirs are primarily distributed in Middle Eastern carbonates, deep-water sandstones, hot dry rock geothermal systems, and deep carbonate/basement rock reservoirs in basins such as Tarim, Sichuan, and Qaidam in China. They universally face extreme environmental challenges: completion tools must withstand conditions >200°C/100 MPa, with significant risks of high-temperature aging and high-pressure deformation in sealing systems; insufficient thermal stability of fracturing fluids leads to a sharp decline in fracture conductivity, and high-temperature well control incidents cause downtime to exceed 30% of total operational time[1–3]. The global industry is seeking breakthroughs through three main directions—intelligent completion, low-carbon stimulation, and high-temperature resistant materials—to address bottlenecks such as "difficulty in fracture initiation, propagation, conductivity maintenance, and safety control"[4–7]. In recent years, significant progress has been made both internationally and domestically in theoretical innovation, process optimization, material development, and field application, forming distinct technical systems[8–11].

In the field of UHT completion technology, core breakthroughs focus on intelligent control systems and high-temperature sealing tools. Leading international companies have developed high-performance wireless control systems: Halliburton's RezConnect™ utilizes magnetic induction communication (2–10 kHz), withstands 204°C, achieves a data transmission rate of 12 kbps, and reduces horizontal well control response time to 22 seconds[3]; Schlumberger's CertaConnex™ wireless control valve integrates piezoelectric-hydraulic actuation, achieving a control delay of <5 seconds under 210°C/140 MPa[12]. Tool temperature resistance has significantly improved; for instance, the Quasar™ packer employs carbon nanotube-reinforced FFKM sealing rings, offering a sealing life >10 years under 210°C/140 MPa[13]; Shape memory alloy packers (NiTiHf-20) achieve 100% setting success rate by adaptively sealing complex wellbores with ±15% diameter variations through phase transformation[14]. Domestically, parallel important progress has been made: Special sealing materials based on tetrafluoroethylene-propylene rubber (FEPM) reinforced with carbon nanotubes have been successfully developed. Combined with a multi-material composite structure (metal support ring + PTFE protection ring + fluor rubber element), a completion packer rated for 232°C/103.4 MPa has been created. Finite element simulation optimizes contact stress (≥110 MPa), and field application in 14 wells showed zero leakage, meeting API-11D1 V0 grade requirements[15–16];

Analysis of the current status of well testing and completion technology in the Sichuan-Chongqing region verifies the regional applicability of UHT tools[17]; The domestically produced 89-type perforator withstands 210°C and 245 MPa pressure, achieves a 100% firing rate, and costs only 50% of imported technology[18].

Regarding UHT reservoir stimulation processes, technological development centers on energy stimulation and dynamic control. The international frontier explores non-hydraulic methods: Plasma Pulse Fracturing (PPF) releases >500 MPa shockwaves in 350°C granite, forming high-density radial fracture networks (>20 fractures/m), increasing permeability to 15mD[19]; Supercritical CO_2 fracturing-geothermal power generation coupling technology (e.g., the US FORGE project) replaces water-based fluids with CO_2, increasing fracture complexity by 40% and reducing water consumption by 90%[10–11]. Domestically, characteristic processes adapted to complex geological conditions have been formed: In the Jizhong Depression Ordovician buried hill (180°C), the "multi-stage injection + temporary plugging and diverting + sand-carrying acid fracturing" process, combined with large-scale physical modeling and numerical simulation, enabled safe high-rate (11 m^3/min) operations, increasing stimulated reservoir volume by 3.5 times and significantly boosting productivity[20]; The "3D Acid Fracturing" concept proposed for deep sour carbonate reservoirs in the Sichuan Basin achieves efficient utilization of long horizontal sections (>1000 m) through intelligent fracture placement and dynamic acid distribution[6, 21]; In the ultra-deep fractured sandstone (>6000 m) of the Tarim Kuqa Depression, activating natural fractures increased gas well absolute open flow potential by 3–5 times[22–23]. Common technologies include degradable fiber temporary plugging and diverting (PLA fiber), enabling balanced stimulation across 12 stages with a plugging breakthrough pressure >500 psi and a production increase rate of 35%[24]; Self-generating acid systems (Therma Stim™/domestic non-hydrochloric acid-based retarded acid) extend acid-etched fracture length to 45 m and prolong dissolution time by 58–74 times through high-temperature hydrolysis of tert-butyl acetate or slow dissolution[25–26].

In the field of UHT working fluids technology, R&D focuses on high-temperature stability and environmental adaptability. Fracturing Fluid Systems: Internationally developed zirconium/titanium-crosslinked HPAM fracturing fluid withstands temperatures up to 232°C, maintaining viscosity >100 cP after 2 hours of shearing[27]; Self-healing nano-fracturing fluid, incorporating thermosensitive SiO_2 nanoparticles, exhibits conductivity decay rate <10% at 220°C[28]. Domestically, carboxymethyl fracturing fluid maintains viscosity >80 mPa·s after shearing at 180°C and 170 s^{-1} for 120 minutes[20]; Tarim Oilfield's polymer/calcium chloride-weighted fracturing fluid (density 1.35 g/cm^3) is rated for 180°C and can reduce wellhead pressure by 10–20 MPa[29]. Acid Systems: Domestic 180°C clean acid (VES system) achieves a viscosity of 30 mPa·s; Hybrid gelling agents, strengthened by hydrophobic association and nanoparticles, maintain viscosity >15 mPa·s at 180°C, with broken residual acid facilitating flowback[26, 30]. Drilling/ Completion Fluids: Sodium formate-weighted systems (density 1.3 g/cm^3) withstand 170°C, costing 40% less than potassium formate systems[31]; Barite-sag-resistant oil-based completion fluid exhibits a settling factor <0.52 after static aging for 10 days at 190°C, ensuring ultra-deep well safety[1].

In summary, global UHT reservoir development technologies are rapidly evolving towards intelligence, low-carbon solutions, and high-strength materials. Domestic technologies hold distinctive advantages in complex geological adaptability and cost control, while international counterparts currently lead in extreme-temperature (>230°C) tools and disruptive processes (e.g., plasma, supercritical CO_2). Future efforts require strengthening fundamental research on high-temperature materials, integrating intelligent dynamic control technologies, and fostering innovation in green, low-carbon processes.

Geological engineering characteristics of Kunteyi block

The Kunteyi Block in the Qaidam Basin represents an ultra-deep, ultra-high pressure, ultra-high temperature bedrock gas reservoir. Its primary gas-bearing formation comprises bedrock lithologies, including gneiss, granitic gneiss, and quartz schist. The K2 well area is dominated by granitic gneiss, with local diorite and gabbro-diorite occurrences in the KX-X well. Reservoir porosity ranges from 2% to 12% (avg. 5.2%), while permeability varies between 0.028 mD and 13.5 mD (avg. 0.22 mD), indicating poor reservoir quality. The reservoir features well-developed conductive fractures. Fracture porosity averages 0.015%, and the pore system consists of fractures combined with matrix porosity. Formation tests from the K2 well reveal a pressure coefficient of 1.63 and a geothermal gradient of 2.82 °C/100m, classifying it as an overpressured, normal-temperature system. However, significant burial depth results in high reservoir temperatures (up to 206°C).

Rock mechanics testing on cuttings, comprising four sets of measurements parallel/perpendicular to bedding plus log-derived inversions, yielded average engineering parameters: Young's modulus (47.6 GPa) and Poisson's ratio (0.24). The relatively high Young's modulus necessitates increased pumping rates to ensure adequate fracture width. The average brittleness index is 0.51, indicating moderate brittleness. In-situ stress analysis shows minimum horizontal stress averaging 137.9 MPa (gradient: 1.94 MPa/100m) and maximum horizontal stress averaging 159.2 MPa (gradient: 2.21 MPa/100m), resulting in a horizontal stress difference of 21.3 MPa. The maximum principal stress orientation is NE-SW. Application of a fracability index model produced a comprehensive chart for the Kunteyi Block, indicating indices ranging from 0.34 to 0.76 (avg. 0.58). This classifies the reservoir as Category II, requiring large-scale stimulation.

In recent years, six wells underwent testing: four achieved commercial gas flow rates of 115,000–264,000 m^3/d. Of which three wells (e.g., K2) were single-zone tests, while three others (e.g., KX-X) employed multi-zone commingled production. Five wells received completion/stimulation treatments (primarily acidizing and acid fracturing), exhibiting fracture pressures of 106 MPa-118 MPa. Operations were characterized by high treating pressures and significant sand placement challenges, with an average proppant concentration of 9.8%.

Reverse engineering design

An innovative reverse-engineering approach was developed for completion-stimulation design. This methodology reverse-engineers required fracture geometry and stimulation intensity based on geological targets and per-well productivity requirements. Optimal stimulation techniques are subsequently selected, followed by tailored completion methods and tool assemblies. The wellbore architecture is ultimately optimized to support the design.

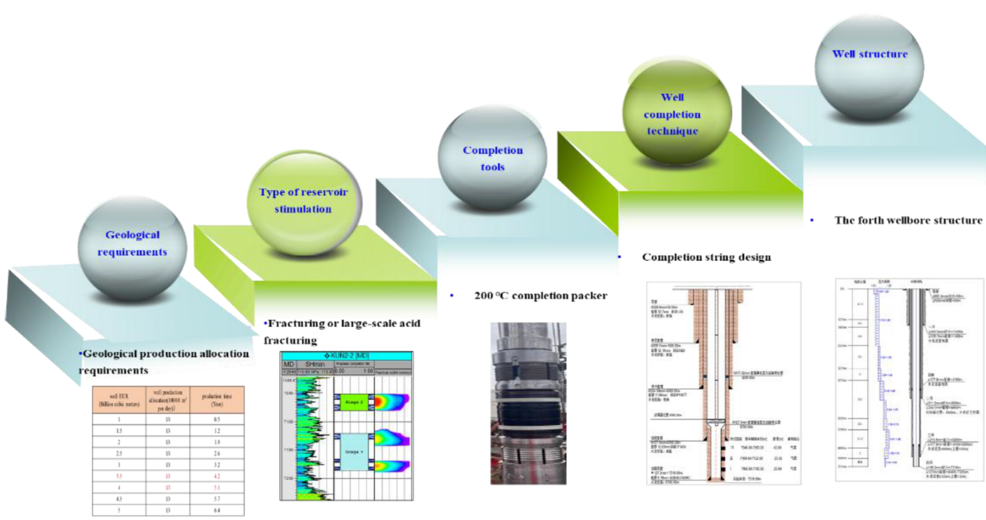

Figure 1—Reverse-engineering methodology for completion-stimulation design

Fracturing design optimization: targeted extended fracture geometry and enhanced well productivity by focusing on nine key parameters (including fracture dimensions and fluid volume). An integrated approach employing backward design and forward optimization was implemented, informed by prior fracturing operations and benchmarking against offset wells. First of all, the fracture dimensions were determined by backward calculation to achieve comprehensive reservoir stimulation. And the perforation strategy and parameters were optimized to establish effective wellbore-reservoir connectivity, ensuring efficient stimulation delivery. The optimal fracture geometry and length required pump rate, fluid volume, and proppant mass optimization. Besides, Required treating pressure was determined considering completion string configuration, wellbore conditions, and fluid friction. Proppant placement was designed to maximize conductivity under safe sand placement constraints. Efficient diversion was employed during multi-layer treatments to ensure effective coverage, while single-layer treatments proceeded without diversion. In the end, the flowback system adhered to dual requirements: (a) initiation only after complete fracture fluid break; (b) controlled flowback post-closure, monitored via wellhead pressure analysis.

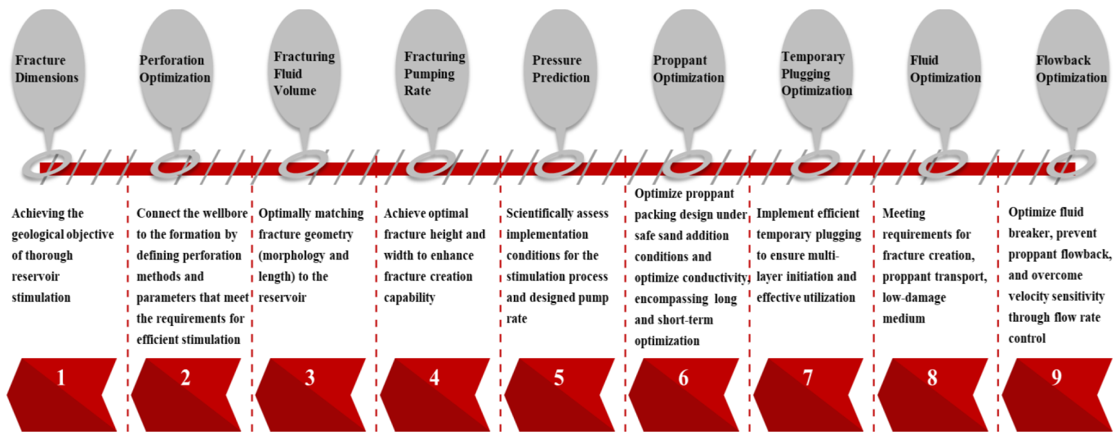

Figure 2—Flowchart of fracturing design optimization

Acid fracturing design optimization: Acid fracturing design targeted extended fracture geometry and enhanced acid-etched conductivity by focusing on eight key parameters (including fracture dimensions and acid volume). An integrated reverse design and forward optimization approach was implemented, leveraging insights from prior acid-fracturing operations and offset well benchmarking.

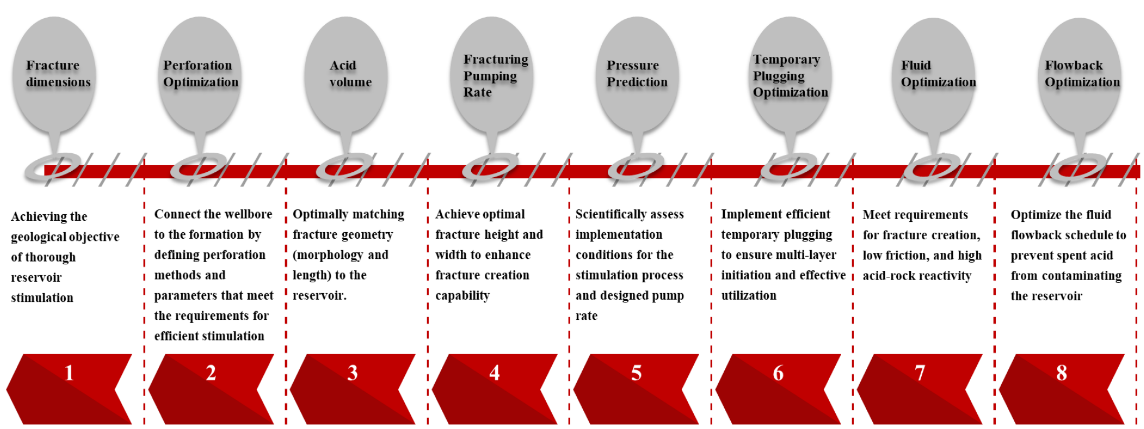

Figure 3—Flowchart of acid fracturing design optimization

Combined Perforation-Testing Completion Technology

Well 1# is the first exploration well in the Kunteyi Block, completed with a 5-spud wellbore structure. The reservoir depth is 7015m, the temperature is 195℃, and the formation pressure is 113.7MPa. In view of the ultra-deep and ultra-high temperature characteristics, the combined perforation-testing technology is adopted. The well has high productivity after testing, and a permanent packer is used for completion. The key tools and parameters are as follows:

Perforating gun

According to the reservoir temperature and pressure characteristics, an 89mm ultra-high temperature and ultra-high pressure perforating gun and SDP35HNS25-4 perforating charges are selected. Through the optimization of the joint sealing structure and the inter-gun support structure, the perforating equipment can withstand a temperature of 210℃ and a pressure of 175MPa.

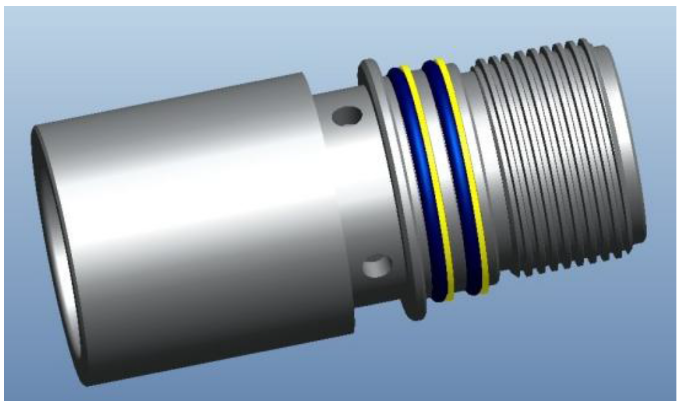

Figure 4—Sealing structure of the perforating gun joint

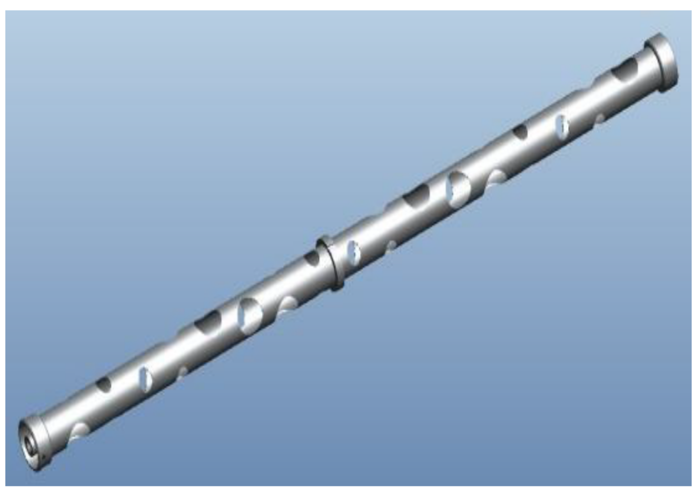

Figure 5—Support structure of perforation guns

Testing tools

A test string with five valves and one packer is used. Compared with the traditional test string, two RD safety circulation valves are adopted to achieve double insurance for circulating kill operations. Considering that the well is ultra-deep and prone to complications during testing, the RTTS packer has been optimally designed. The wall thickness of the mandrel is increased by 3mm, and the tensile strength is improved from 384.2kN to 637kN. The well temperature exceeds 200℃, and the extreme high temperature imposes high requirements on the temperature resistance of seals. Therefore, Chemaraz526 series sealing rings are used, which can withstand a temperature of 232℃ and have anti-inner explosion performance.

Figure 6—Diagram of Ultra Deep and Ultra High Temperature Testing Tools

Table 1—Comparison and statistical table of RTTS packer before and after improvement

Thickness (mm)		Tensile Strength (kN)	
Before Improvement	After Improvement	Before Improvement	After Improvement
4.85	7.85	384.2	637

Completion tools

According to the wellbore structure, pressure and temperature resistance requirements, and production demand, a 7-inch THT packer with a pressure resistance of 15,000 psi and a temperature resistance of 232℃ is selected.

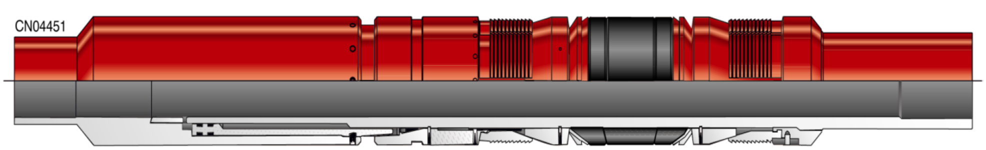

Figure 7—The ultra-deep and ultra-high temperature testing tools

Table 2—The parameter of THT hydraulic permanent packer

Name	TNT Completion Packer
Casing Size	7'/177.8mm
Applicable Weight	35-38 #(152.5-150.4mm)
Maximum Outer Diameter	5.750 inch/146.05mm
Minimum Inner Diameter	2.897inch /73.58mm
Material	9Cr1Mo
Minimum Yield Strength	80000 psi/551.58MPa
O-Ring & Elastomer Element Material	AFLAS RUBBER
Connection Type	3-1/2''9.20VAMTOP B X P
Temperature Rating	100-450F/37.78-232.22°C
Service Condition	H2S/CO2
Pressure Rating	15000psi/103.4MPa
Internal Yield Pressure	10000psi/68.95MPa
External Yield Pressure	10000psi/68.95MPa
Tensile Strength	227000pound/1030.6KN

Integrated Perforation-Acidizing Completion Technology

For Well #2, located at a reservoir depth of 7,310 meters with a reservoir temperature of 205.4°C, the first-stage completion technology was employed (using the same completion tools as in the first-stage perforation-testing integrated completion technology). Following perforation, THT packers were utilized for completion; however, no natural productivity was observed. Consequently, a small-scale acidizing stimulation technique was applied to further enhance connectivity between the wellbore and the reservoir, eliminate and break through near-wellbore reservoir pollution, and restore or improve gas production capacity in the perforated interval. Nevertheless, the small-scale acidizing yielded insignificant results and failed to meet geological targets. A new exploration and evaluation well were thus deployed in the Kunteyi Block, but its perforation also did not achieve the intended geological objectives. To meet the daily production requirement of 130,000 units, further optimizations were made: the completion string was refined, a large-diameter completion string was adopted, the acid formula was upgraded and optimized, and large-scale acid fracturing operations were conducted—measures that significantly boosted single-well production.

String Design

Well #2 adopted completion string for acidizing. The string combination is: gas-tight tubing 3-1/2" (6200m) + 2-7/8" (700m), with material BG110 and thread type: BGT2 gas-tight thread.

1. Packer setting position: 6218.18m (avoiding casing collars)
2. Downhole tools run into the well are acid-resistant and temperature-resistant, with temperature resistance not lower than 200°C.
3. Tubing length configuration: 88.9×9.52mm (2500m), 88.9×6.45mm (3700m), 73.0×5.51mm (700m).

Well #3 adopted completion string for acid fracturing. The string combination is: 4-1/2" (4280m) + 2-7/8" (2150m), with materials HS110 and P110, and completion packers were used for setting.

1. Packer setting position: 6550m (avoiding casing collars)

2. Downhole tools run into the well are acid-resistant and temperature-resistant, with temperature resistance not lower than 200℃.
3. Tubing length configuration: 114.3×12.7mm (1380m) + 114.3×9.65mm (1500m) + 114.3×8.56mm (1400m) + 88.9×9.52mm (2150m)

Ultra-High Temperature Acid System

The dissolution rate of cuttings increases with the increase of acid concentration, but the increase rate of dissolution rate slows down significantly when the HCl concentration exceeds 10%. Therefore, the recommended HCl concentration is 10%. In the experiment where the acid was kept at 200℃ for 6 hours, no pitting corrosion occurred on the steel sheets, and the corrosion rate was only 46.3 g/(m²•h), meeting the requirements of standard indicators. In addition, this acid system has good compatibility, with no precipitation, flocculation, or other phenomena. Considering the mud pollution, the concentration of hydrochloric acid in the pre-treatment acid is appropriately increased to 12%.

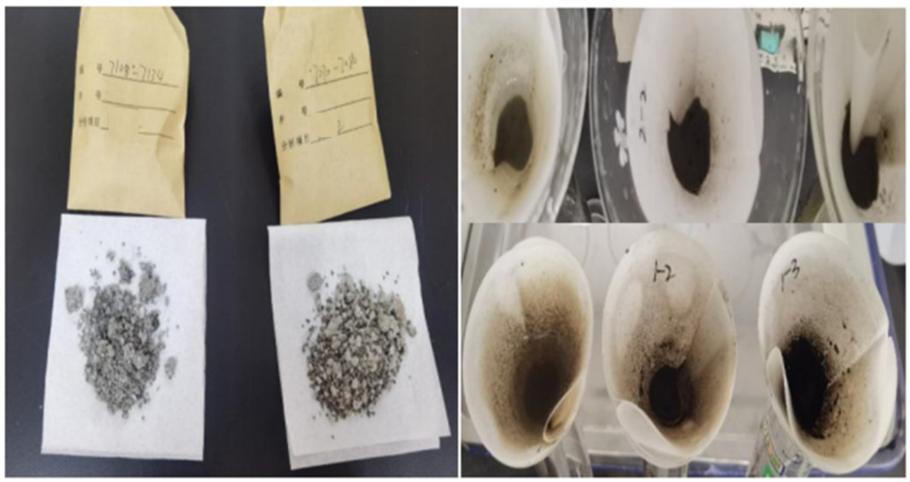

Figure 8—Experiment on acid dissolution of cuttings

Table 3—Acid Performance Test Table

Test Acid Formula	Standard Requirements	Dynamic Corrosion Rate, g/(m²•h)	Standard, g/(m²•h)
20%HCl + 5.0% corrosion inhibitor + 0.2~0.5% synergist (enhanced type)	200℃、6h	55.6	180℃、≤70
12%HCl + 3%HF + 5.0% corrosion inhibitor + 0.2~0.5% synergist (enhanced type)	200℃、6h	46.3	180℃、≤70

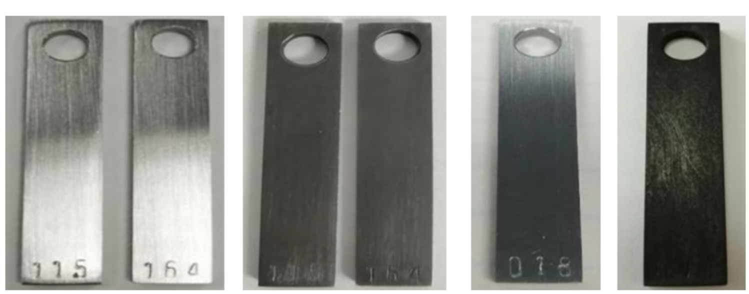

Figure 9—6h Corrosion rate test under 200# experimental conditions

During the application process, this technology exposed some problems: first, after running the stimulation string into the mud, there is a risk of mud precipitation in the reservoir section and coiled tubing getting stuck or buried when using coiled tubing to displace the mud. Second, the stimulation string can only perform single-layer acidizing/multi-layer combined fracturing, which cannot meet the requirements of separate-layer stimulation. Third, the acidizing string cannot achieve isolation between the production tubing and casing, which restricts subsequent underreaming or easily leads to high casing pressure.

Well3# Construction Parameters

For the construction parameters of Well #3, please refer to the Case Application in this article.

Shallow-Tubing Stimulation-Completion

well 4# located at a reservoir depth of 7,310 meters with a reservoir temperature of 182℃. To explore the adaptability of the sand fracturing technology and further increase the single-well production, Well 4# adopts the completion method of shallowly setting Y531 packers with tubing for stimulation.

String design

1. Large-Bore Tubing String Configuration:114.3*12.70mm（1447m）+ 114.3*9.65mm（1430m）+ 114.3*8.56mm （956m）
2. Mechanical verification: This string combination can meet the construction displacement of 6.0m^3/min, with an estimated pump pressure of 101MPa and a required balanced casing pressure of 34-46MPa. The minimum safety factor of the string is 1.520
3. Fracturing wellhead: Model KL78/65-140P

Y531 Packer

Well 4# adopted the Y531 packer, which can withstand a temperature of 200°C and a pressure difference of 100 MPa, and allows for lifting to unset after operation.

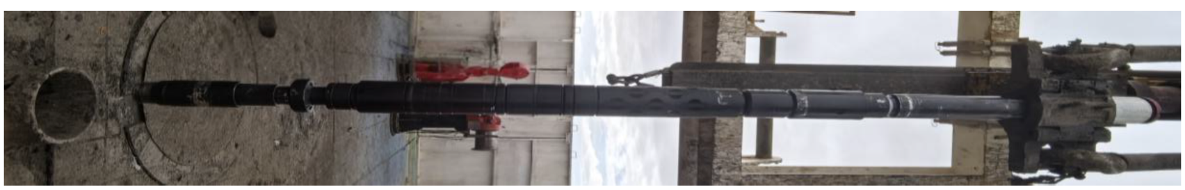

Figure 10—Y531 Packer

Table 4—Key parameters of Y531 Packer

Technical Parameters	Specifications
Overall Length	1485mm±10mm
Casing OD	148mm
Bore Diameter	60mm
Ball Seat Bore	50.5mm
Ball Seat OD	55mm
Operating Temperature	200°C
Differential Pressure Rating	100Mpa
Release Mechanism	Pressure-release auto-release
Connection Type	Upper: 4-1/2TBG；Lower: 3-1/2UPTBG

Fracturing Parameters

The large-scale temporary plugging fracturing treatment of Well 4# has been completed, with a total fluid volume of 1644.4m^3, a maximum construction pressure of 116.7MPa, a maximum displacement of 5.3m^3/min, a sand volume of 75.1m^3, and a maximum sand ratio of 18%.

Cased-Hole Staged Fracturing Technology

Well 5# located at a reservoir depth of 7,344 meters with a reservoir temperature of 197°C. To further increase the construction scale, expand the stimulated volume, and achieve synchronous production from multiple layers, this well explores the adaptability of the casing segmented fracturing technology, providing on-site tests for the profitable development of the Kunteyi Block.

Completion tools

According to the prediction formula for surface construction pressure, when the surface construction pressure is 116 MPa, the cement sheath with a density of 1.0 g/cm^3 can basically balance the casing fluid column pressure. It is predicted that the pressure at the casing hanger position (4700 m) will be 113.41 MPa. The pressure difference resistance of the hanger has been increased from 105 MPa to 140 MPa, and its performance meets the requirements of multiple fracturing operations.

Fracturing tools

Geological experts have optimally selected four main production layers. To achieve the full utilization of these four layers, cable-pumped bridge plugs are used to isolate the already stimulated layers, thus enabling separate stimulation of each of the four layers. The bridge plug can withstand a temperature of 191°C and a pressure difference of 15K psi.

Figure 11—Staged bridge plug

Table 5—Parameters of bridge plug

Casing Specs				Plug Specs						O/R
O.D. inch (mm)	Weight Range lb/ft (kg/m)	Min I.D. inch (mm)	Max I.D. inch (mm)	O.D. inch (mm)	Ball Drop Min I.D. inch (mm)	Flo-back Min. inch (mm)	Length inch (mm)	Ball size inch (mm)	Setting Tool	High Temp/High PSI
5-1/2 (139.7)	26.00-28.40 (38.7-42.3)	4.44 (112.8)	4.55 (115.5)	4.13 (104.80)	1.25 (31.8)	0.63 (16.0)	27.50 (698.5)	2.13 (54.0)	Magnum "A-1", Baker #20 or Owen	375℉ (191℃) 15K psi (103.4MPa) Composite/FKM Elastomer
	23.00 (34.2)	4.67 (118.6)	4.67 (118.6)	4.25 (107.9)						

Fracturing fluid

An ultra-high temperature fracturing fluid resistant to 230°C was used, and over 100 sets of laboratory experiments were conducted to optimize the best fracturing fluid formula. Evaluations were carried out on the performance of the fracturing fluid such as temperature and shear resistance, sand-carrying capacity, and gel breaking property, which meet the fracturing requirements of reservoirs with a temperature of 200°C. According to the analysis of the construction curve, the fluid drag reduction rate is over 74%.

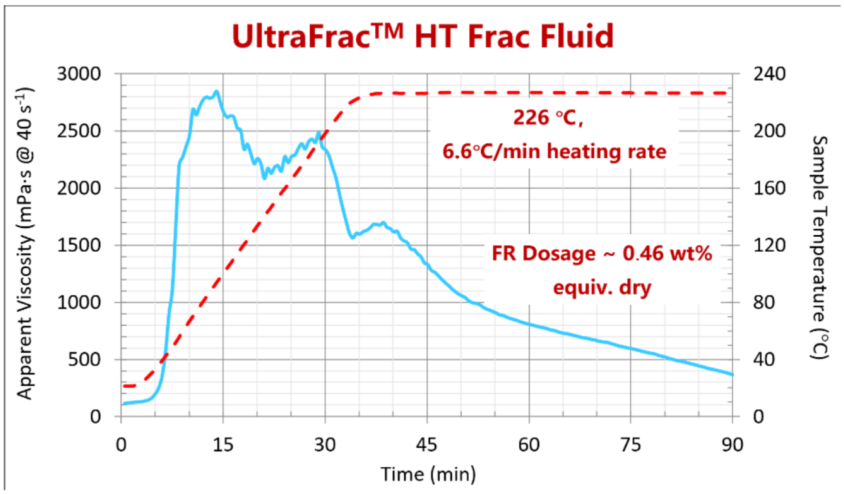

Figure 12—Experimental results of fracturing fluid temperature and shear resistance

Treatment parameters

The large-scale fracturing treatment of Well 3# has been completed, with a total fluid volume of 1872.0m^3, a maximum construction pressure of 100.7MPa, a maximum displacement of 10m^3/min, a sand volume of 94.5m^3, and a maximum sand ratio of 19%. By utilizing casing fracturing technology combined with sweet spot identification techniques, the pumping rate was increased from 6 m^3/min (the maximum rate recorded in offset wells) to 10 m^3/min. This optimization also resulted in a main treatment pressure of 95.7 MPa, representing an 13% reduction compared to the 110 MPa observed in adjacent wells.

Table 6—The Treatment parameters

Treatment parameters															
Bridge plug position, m	cluster	pump rate, m^3/min	breakdown pressure, MPa	max treatment pressure, MPa	fractring volume					sand volume			maximum sand ratio	average sand ratio	shutdown pumping pressure, MPa
					total fractring volume, m^3	net fractring volume, m^3	Acid, m^3	fracturing liquid, m^3	Slickwater, m^3	total sand volume	70-140, m^3	40-70, m^3			
7121	4	8-10	95.7	100.7	1872	1777.5	100	1592.5	85	94.5	82	12.5	19	11.6	88.0

In the application process, this technology has exposed some problems. Firstly, it is difficult to run bridge plugs in highly deviated wells, and the running time is long. secondly, it is challenging to drill and mill bridge plugs, especially blind bridge plug.

Case application

Acid fracturing treatment

The acid fracturing operation in Well 3# achieved a maximum treating pressure of 107.8 MPa with a pump rate of 5.0–6.6 m^3/min. A total fluid volume of 1,304.2 m^3 was injected, including 726 m^3 of 230°C high-temperature acid. The relatively low shut-in pressure of 61 MPa and pronounced water hammer effect indicate effective stimulation. The maximum daily gas production is 26 4000m^3/d.

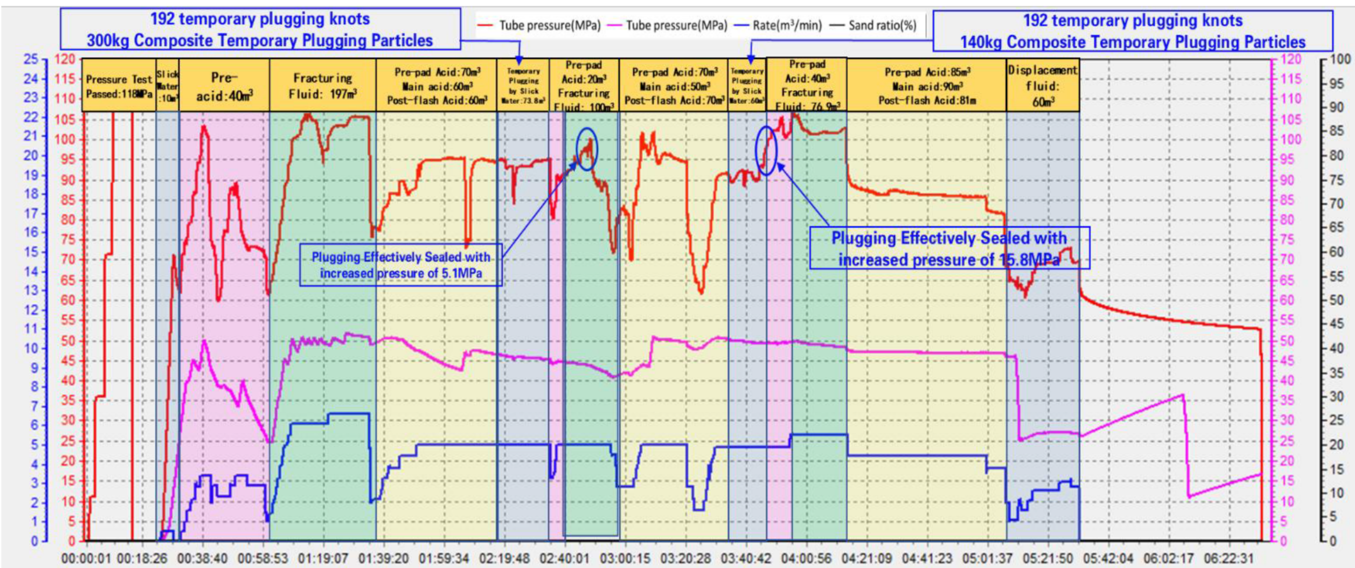

Figure 13—Treatment curve of KX-X well

Analysis of fracture initiation

Fracture initiation characteristics were characterized using pressure wave cepstral analysis of millisecond pressure fluctuations derived from high-frequency pressure monitoring data. Comprehensive analysis showed that a total of 5 perforation clusters were activated during the acid fracturing stimulation of this well, corresponding to the formation layers: III-1, III-2, III-3, II, and I-2, with a perforation activation rate of 83%. Based on the analysis of fracture initiation time to evaluate the fluid intake of each perforation cluster, the main fluid-intaking clusters before temporary plugging were III-2 and III-3; after the first temporary plugging, the fluid-intaking clusters were III-1 and II; and after the second temporary plugging, the main fluid-intaking clusters were II and I-2.

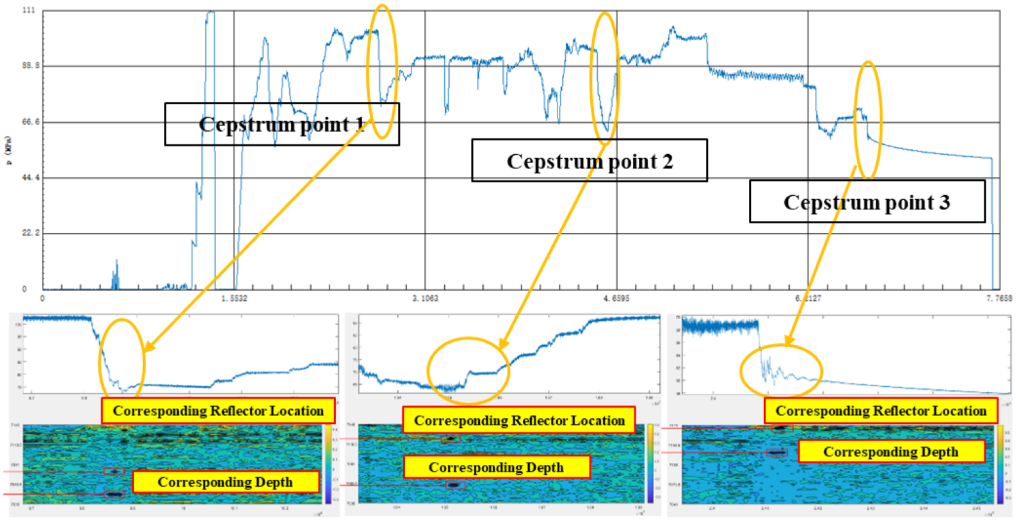

Figure 14—Analysis of fracture initiation before and after temporary plugging

Analysis of liquid inflow in each cluster

Based on the fluid injection volume analysis across individual clusters in the acid fracturing operation, perforation clusters with superior reservoir properties and relatively lower stress preferentially accept fluid. The temporary plugging demonstrates remarkable effectiveness—initiating new fractures and ensuring uniform fluid distribution. Thus, the temporary plugging diversion technique proves effective in stimulating ultra-deep reservoirs.

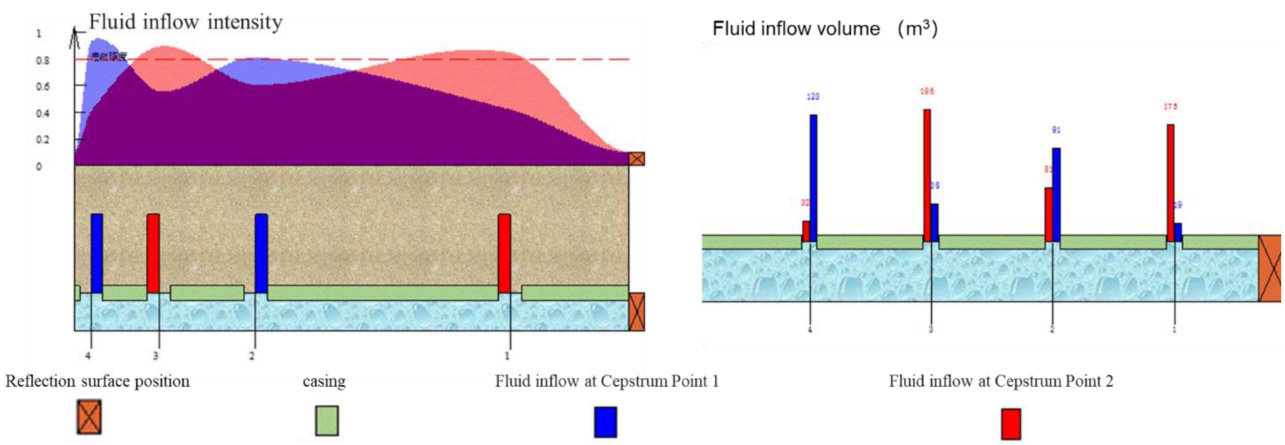

Figure 15—Liquid inflow volume in each cluster (a) Liquid inflow volume curve (b) Quantitative fluid inflow volume analysis chart

Analysis of pressure drop test

A log-log plot analysis was performed on the pressure, pressure derivative and time during the pressure drop phase. The results show that there is an obvious linear flow straight segment in the derivative plot, indicating that a main fracture was formed. After the end of linear flow, the slope of the derivative changes, indicating the formation of a secondary fracture network, with a fracture shape factor of 0.73. Through G-function analysis, it was found that no fracture closure occurred during the shut-in test.

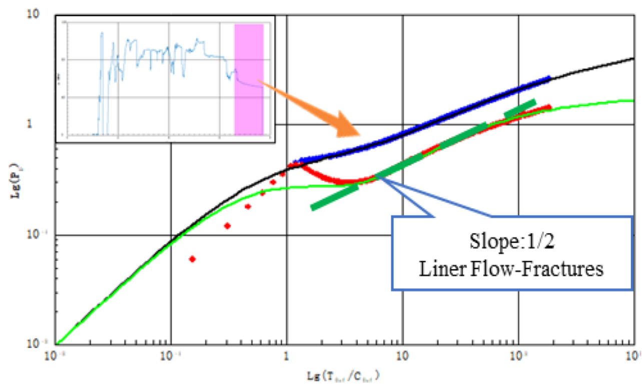

Figure 16—Curve of pressure drop test analyzing

Analysis of fracture morphology

Based on the acid fracturing numerical simulation analysis, the effective acid-etched fracture length is 116m, with a fracture conductivity of 10.24 D·cm. Other key parameters are summarized in the table below.

Table 7—The parameter of fracture morphology

Liquid volume injected（m^3）	1304.19
Number of perforation clusters	6
Number of activated clusters	5
Initial pressure（MPa）	111.36
Average pressure （MPa）	115.88
Permeability of SRV area（md）	8.82
Fracture length（m）	116.99
Sweep length（m）	188.03
Fracture height （m）	20.59
Fracture conductivity（D·cm）	10.24
Main SRV （10000m^3）	7.49
Total SRV（10000m^3）	33.78
Maximum liquid production rate （m^3/d）	21.61
Fracture network morphology factor	0.73
Fracture Initiation Ratio	83%

Conclusions

Addressing the completion-stimulation challenges in ultra-deep bedrock reservoirs (200°C), this study pioneers a reverse-engineered design with forward-execution methodology. Four integrated technologies, including Combined Perforation-Testing Completion, Integrated Perforation-Acidizing Completion, Shallow-Tubing Stimulation-Completion, Cased-Hole Staged Fracturing, were developed and systematically validated. Critical technical specifications, including tool configurations, stimulation fluids, and operational parameters, were quantified for each technology, with implementation challenges identified. This work establishes actionable benchmarks for Optimized technique selection in ultra-deep and ultra-HTHP reservoirs, engineering-driven tool design, high-temperature fluid systems, parametric operation frameworks. These advancements provide a methodological foundation for safe, efficient completion-stimulation operations in extreme geothermal environments.

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